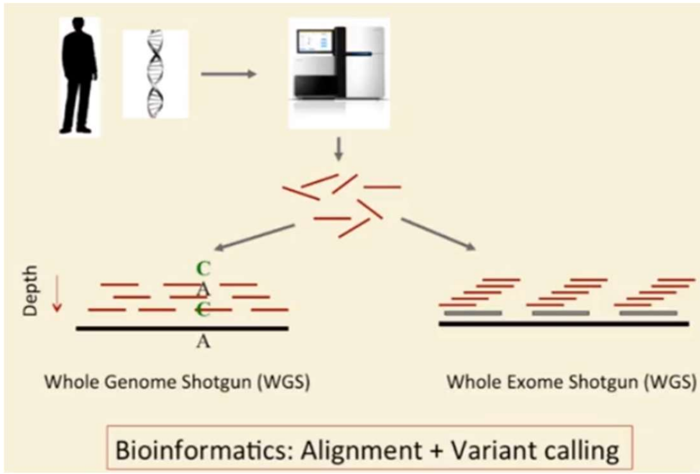


SAM Format Review

141217_CIDR4_0073_BHCFG7ADXX:2:1111:3128:29074	Read id
99	**FLAG**
chr1	Chr
10021	Start
0	Mapping quality
50M	**CIGAR** (alignment)
=	Mate chr
10151	Mate start
180	Mate dist
ACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAAC	Query seq
@DC?=2.FFGE@7>C62>BGABGB9HFBAFIIHEGFIIIHFAIIGDA<FC	Query base quals
	Alignment score
NM:i:0	Edit distance to reference
NH:i:10	Number of hits
XS:A:-	Strand
HI:i:0	Hit index for this alignment

Tags: [A-Za-z][A-Za-z]:[AifZH]:.*
where A =character; i = integer; f = float; Z=string; H = hex string

(SAMtools) mpileup Format

```
#CHR  POS     REF NRDS BASES                  QUALITIES
17   7574014  A   19    ..,,,,.,,..,.,,,,,.,  CdA?@C@C6CBCCAB=BBA
#(optional)READPOS
74,72,68,65,61,58,67,55,53,49,48,40,35,34,30,22,32,14,12,8
```

- . match, forward
- , match, reverse
- T mismatch (to T), forward
- t mismatch (toT), reverse
- ^ beginning of read
- $ end of read
- +[0-9]+[ACGTNacgtn]+ insertion in the reference (e.g., +3ACC)
- -[0-9]+[ACGTNacgtn]+ deletion from the reference (e.g., -2GG)
- > reference skip

VCF (Variant Call Format) and BCF (Binary Call Format)

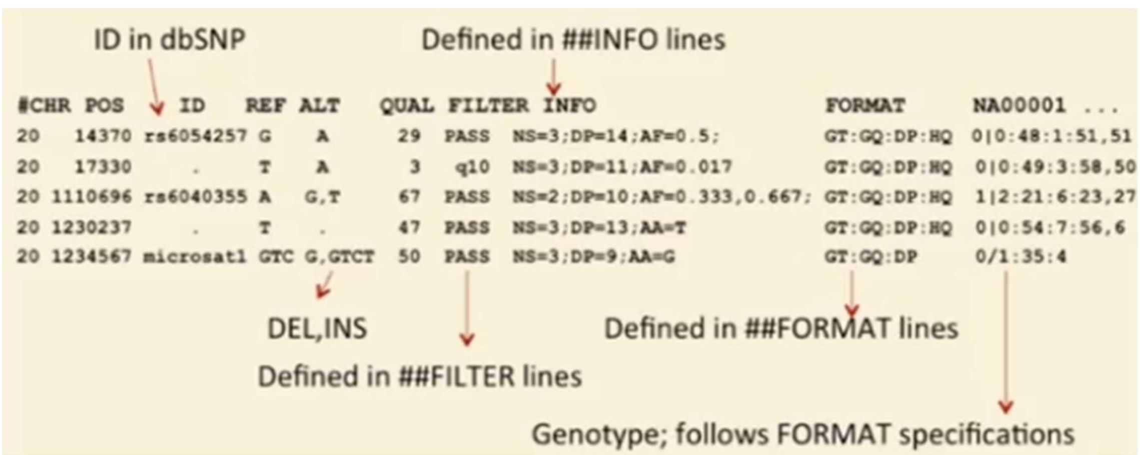
```
##fileformat=VCFv4.2
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">

#CHROM POS     ID        REF ALT    QUAL FILTER INFO                              FORMAT      NA00001 ...
20     14370   rs6054257 G   A      29   PASS   NS=3;DP=14;AF=0.5;                GT:GQ:DP:HQ 0|0:48:1:51,51
20     17330   .         T   A      3    q10    NS=3;DP=11;AF=0.017               GT:GQ:DP:HQ 0|0:49:3:58,50
20     1110696 rs6040355 A   G,T    67   PASS   NS=2;DP=10;AF=0.333,0.667;        GT:GQ:DP:HQ 1|2:21:6:23,27
20     1230237 .         T   .      47   PASS   NS=3;DP=13;AA=T                   GT:GQ:DP:HQ 0|0:54:7:56,6
20     1234567 microsat1 GTC G,GTCT 50   PASS   NS=3;DP=9;AA=G                    GT:GQ:DP    0/1:35:4
```

INFO

FILTER

FORMAT



VCF: INFO

```
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
```

Other standard fields:

1000G, AA (ancestral allele), CIGAR, MQ0 (# reads with MAPQ==0), etc.

Examples

```
NS=3;DP=14;AF=0.5;
NS=3;DP=11;AF=0.017;
NS=2;DP=10;AF=0.333,0.667;
NS=3;DP=13;AA=T
NS=3;DP=9;AA=G
```

VCF: FORMAT

```
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype
Quality">
```

Other standard fields:

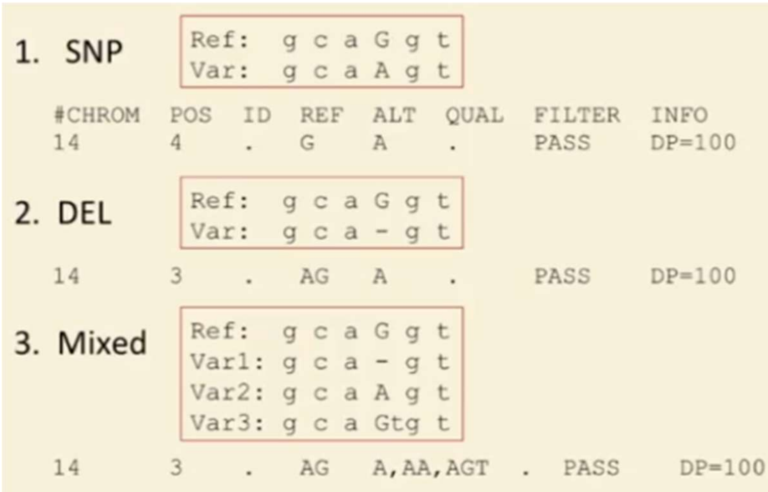
FT (filter pass), GL (genotype likelihood), MQ (mapping quality), etc.

Example

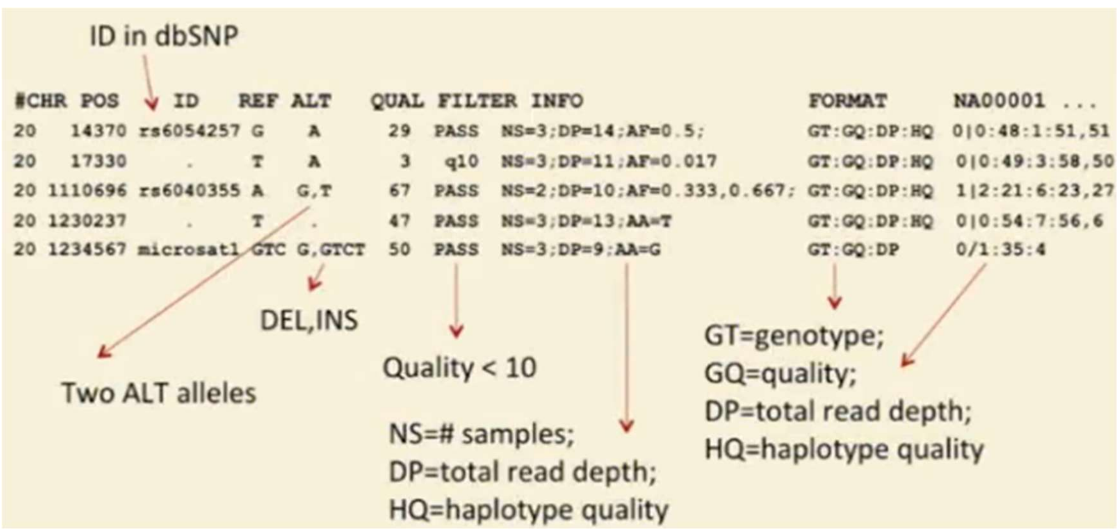
```
FORMAT          NA00001 ...
GT:GQ:DP:HQ     0|0:48:1:51,51
GT:GQ:DP:HQ     0|0:49:3:58,50
GT:GQ:DP:HQ     1|2:21:6:23,27
GT:GQ:DP:HQ     0|0:54:7:56,6
GT:GQ:DP        0/1:35:4
```

0 allele is present in the two samples, quality is 48, one read mapping, the haplotype quality (2 values) is 51 and 51

VCF: Entries



Decoding VCF Entries



Alignment in Unix

- Fast tools for mapping large number of sequences by using compressed version of genome as an index; look for contiguous matches with small number of indels
- **Bowtie**
    - bowtie-build reference.fasta index_base
    - bowtie index_base reads.fq output.sam
    - Options
        - -q: input is FASTQ (default)
        - -f: input is FASTA
        - -p <int>: number of processor threads to use
        - --sam: output in SAM format
        - -v <int>: number of mismatches allowed in alignment
        - -m <int>: suppress reads with more than <int> valid alignments
    - Paired-end reads: bowtie -1 reads_1.fq -2 reads_2.fq index_base output.sam
- **BWA**
    - bwa index reference.fasta
    - Options
        - -t <int>: number of threads
        - -M: mark shorter split hits as secondary (for compatibility with Picard/GATK)
        - -R <string>: specify read group header line in SAM output
    - Short reads: bwa aln reference.fasta reads.fq > reads.sai
bwa samse reference.fasta reads.sai reads.fq > output.sam
    - Paired-end reads: bwa aln reference.fasta reads_1.fq > reads_1.sai
bwa aln reference.fasta reads_2.fq > reads_2.sai
bwa sampe reference.fasta reads_1.sai reads_2.sai reads_1.fq reads_2.fq > output.sam